



CSE 4531: E-Commerce and Web Security

Quiz: 1

Duration: 30 minutes

Total Marks: 15

Date: 22.12.25

Answer all three (3) questions. Figures inside the boxes in the right margin indicate marks of each question.

Student ID:

1. For each of the following security goals—**Confidentiality**, **Integrity**, and **Availability**—do the following: 3
 1. Describe a **realistic scenario** in which the security goal is compromised in an e-commerce system.
 2. Explain the **technical mechanism or type of attack** that leads to the compromise.
 3. Discuss the **impact** of the compromise on the system, stored data, and end users.
2. The ride-sharing application provides drivers with a profile update form through which they can modify **their own vehicle description** (illustrated in Figure 1.1). 6

Update Vehicle Information

Vehicle Description:

e.g., White Toyota Corolla, 2019 model

Submit

Figure 1.1: Profile update form used by drivers to edit vehicle details (entry point for user input).

When the form is submitted, the backend executes the following SQL query:

```
UPDATE drivers
SET vehicle_info = '$vehicle'
WHERE driver_id = $did;
```

Where:

- **\$vehicle** contains the vehicle description entered by the driver through the form.
- **\$did** is the authenticated driver's ID obtained from the session.

Only the **vehicle_info** field is intended to be editable through this form.

The **drivers** table contains a column named **rating**, which drivers are **not authorized** to modify. For example, **Alice**, a legitimate driver, currently has a rating of **4.8**.

Assume that:

- User input is **not sanitized**.
- Prepared statements are **not used**.
- **Eve**, a malicious driver, knows Alice's `driver_id` which is `"DID042"`.

Eve intends to **reduce Alice's rating from 4.8 to 1.2**. Demonstrate an SQL injection attack by specifying the input **Eve could enter in the Vehicle Description field** to modify Alice's rating.

Your answer must clearly specify the **injected payload** (5)

3. A trusted e-commerce website allows customers to submit product reviews. These reviews are stored in the database and later displayed on product pages **exactly as submitted**.

Alice, a legitimate customer, is authenticated while browsing product pages. Eve, a malicious user, notices that **JavaScript code included in a product review is executed in the browsers of users who view the affected product page**.

Answer the following:

(a) Injection and Storage of Malicious Code (2.5 marks)

- Explain how Eve can inject malicious code.
- Explain why the application stores the malicious code.

(b) Execution of Malicious Script in Alice's Browser (2 marks)

- Describe, step by step, how the injected script executes in Alice's browser.

(c) Session Capture (1.5 marks)

- Explain how Alice's session information can be captured.